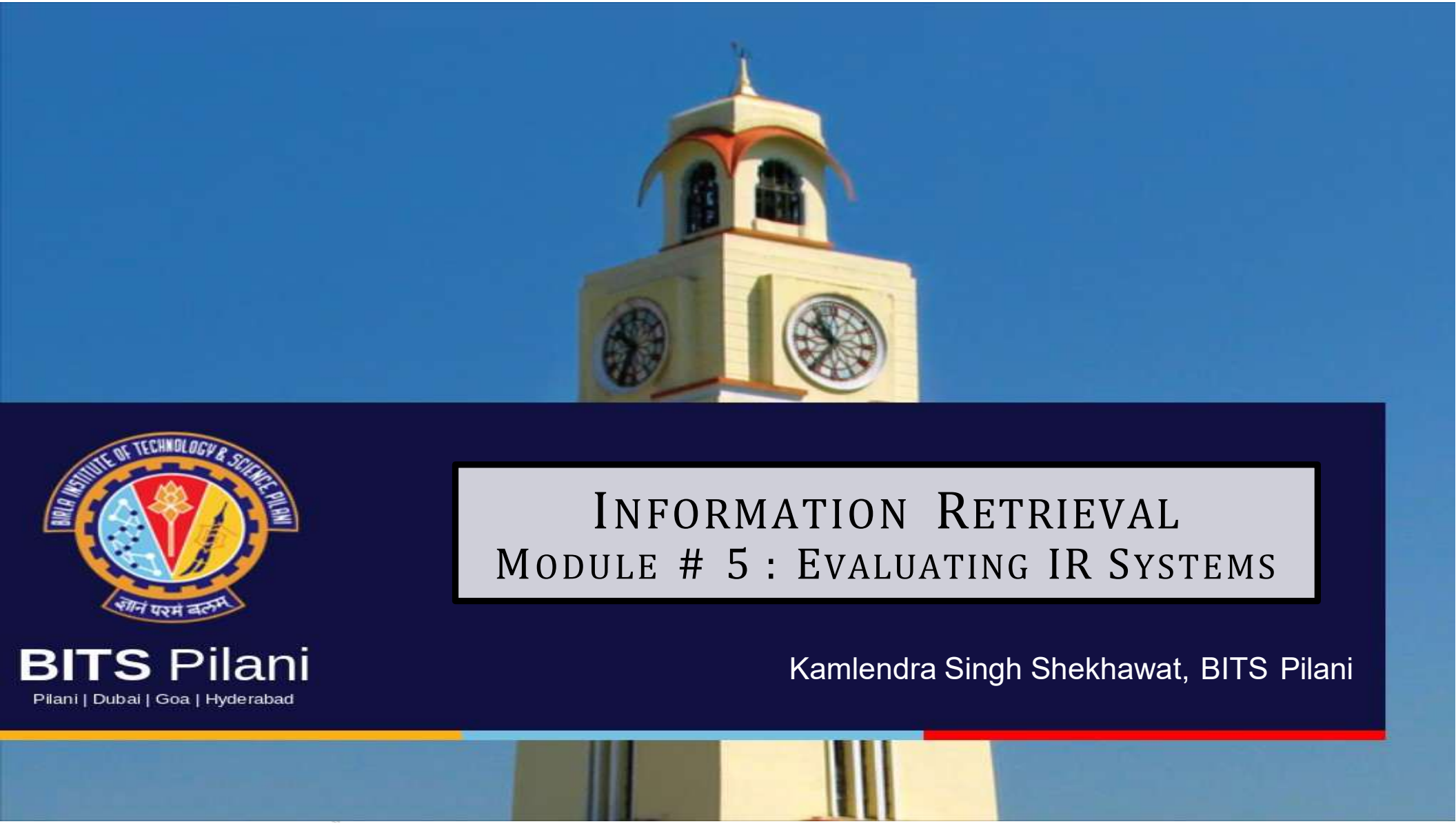

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INFORMATION RETRIEVAL
MODULE # 5 : EVALUATING IR SYSTEMS

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The instructor is gratefully acknowledging the authors who made their course materials freely available online.

This deck is prepared by Kamlendra Singh Shekhawat.

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5. Non-Binary Relevance

Module Topics....

- Evaluation in Information Retrieval
- Evaluation in unranked retrieval sets
- Evaluation in ranked retrieval sets

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SITUATION

Thanks to your stellar performance in IR Course, you quickly rise to VP of Search at internet retail giant IRSearch.com. A nerd X, claims to have built a better search engine. Do you

- Counsel X to take IR Course?
- Try a few queries on the search engine developed by X?

What could you ask X?

- How fast does it index?
  - Number of documents/hour
  - Incremental indexing – IRSearch adds 10K products/day
- How fast does it search?
  - Latency and CPU needs for IRSearch 5 million products
- Does it recommend related products?
- This is all good, but it says nothing about the quality of X's search system
- You want IRSearch's users to be happy with the search experience

How do you tell if users are happy?

- Search returns products relevant to users
  - How do you assess this at scale?
- Search results get clicked a lot
  - Misleading titles/summaries can cause users to click
- Users buy after using the search engine
- Repeat visitors/buyers
- Do users leave soon after searching?
- Do they come back within a week/month/. . . ?

Happiness: Elusive to Measure

- Most common proxy: relevance of search results
- Pioneered by Cyril Cleverdon in the Cranfield Experiments
- But how do you measure relevance?

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Measuring Relevance

Three elements:

1. A benchmark document collection
2. A benchmark suite of queries
3. An assessment of either Relevant or Non-relevant for each query and each document

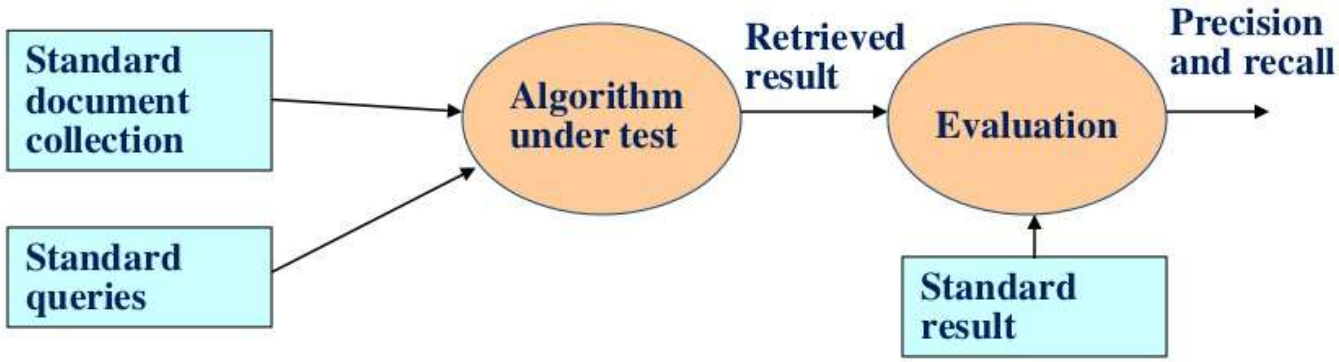
BENCHMARKS

A benchmark collection contains:

- A set of standard documents and queries/topics.
- A list of relevant documents for each query.

Standard collections for traditional IR:

- Smart collection: ftp://ftp.cs.cornell.edu/pub/smart
- TREC: http://trec.nist.gov/



The TREC Benchmark

- TREC: Text REtrieval Conference (http://trec.nist.gov/)
- Originated from the TIPSTER program sponsored by Defense Advanced Research Projects Agency (DARPA).
- Became an annual conference in 1992, co-sponsored by the National Institute of Standards and Technology (NIST) and DARPA.
- Participants submit the P/R values for the final document and query corpus and present their results at the conference.

TREC TASKS

- Ad hoc: New questions are being asked on a static set of data.
- Routing: Same questions are being asked, but new information is being searched. (news clipping, library profiling).
- New tasks added after TREC 5 - Interactive, multilingual, natural language, multiple database merging, filtering, very large corpus (20 GB, 7.5 million documents), question answering.

Characteristics of the TREC Collection

- Both long and short documents (from a few hundred to over one thousand unique terms in a document).
- Test documents consist of:

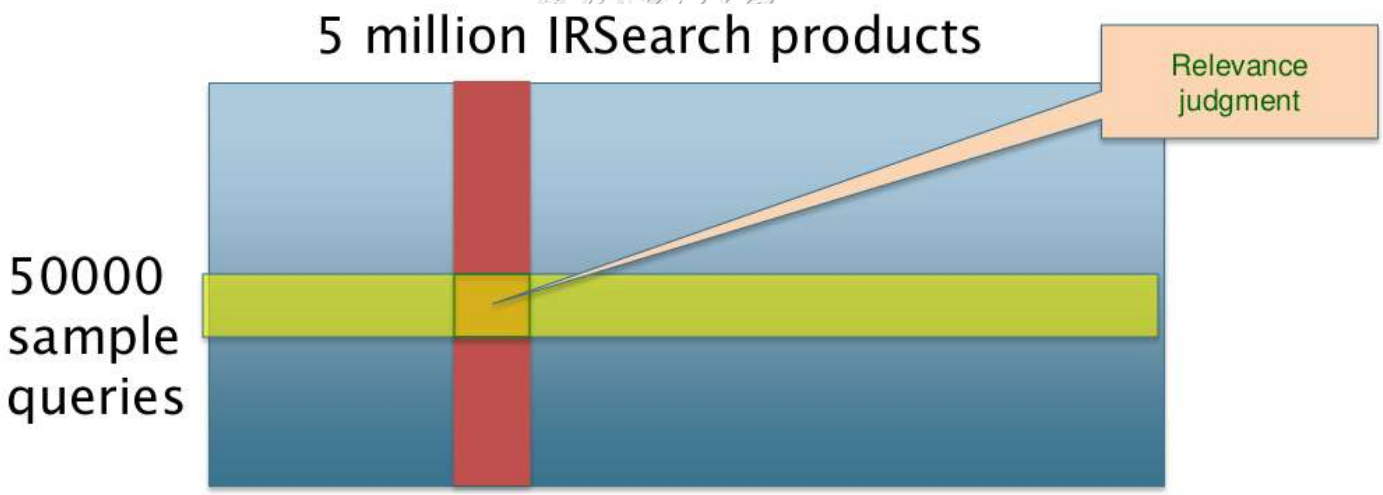
Abbreviation	Source	Size (MB)
WSJ	Wall Street Journal articles (1986-1992)	550
AP	Associate Press Newswire (1989)	514
ZIFF	Computer Select Disks (Ziff-Davis Publishing)	493
FR	Federal Register	469
DOE	Abstracts from Department of Energy reports	190

Standard Test Collections

- TREC
- Cranfield collection – small
- GOV2 web page collection
- NII Test Collections for IR Systems – East Asian language and cross-language information retrieval
- Cross Language Evaluation Forum (CLEF – European languages and cross-language information retrieval
- Reuters-21578 and Reuters-RCV1 – text classification
- 20 Newsgroups – text classification

How to measure the quality of a new search algorithm?

- Benchmark documents
- Benchmark query suite
- Judgments of document relevance for each query



Relevance Judgments

- Binary (relevant vs. non-relevant) in the simplest case
- More nuanced relevance levels also used(0, 1, 2, 3. . .)
- What are some issues already?
  - 5 million times 50K takes us into the range of a quarter trillion judgments
  - If each judgment took a human 2.5 seconds, we'd still need 10^{11} seconds, or nearly $300 million if you pay people $10 per hour to assess
  - 10K new products per day

Crowd Source Relevance Judgments?

- Present query-document pairs to low-cost labor on online crowd-sourcing platforms
- Hope that this is cheaper than hiring qualified assessors
- Lots of literature on using crowd-sourcing for such tasks
- You get fairly good signal, but the variance in the resulting judgments is quite high

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Evaluating an IR system

- Relevance is assessed relative to the user need, not the query.
- E.g., Information need: My swimming pool bottom is becoming black and needs to be cleaned.
- Query: pool cleaner
- Assess whether the doc addresses the underlying need, not whether it has these words

Difficulties in Evaluating IR Systems

- Effectiveness is related to the relevancy of retrieved items.
- Relevancy is not typically binary but continuous.
- Relevancy, from a human standpoint, is:
  - Subjective: Depends upon a specific user's judgment.
  - Situational: Relates to user's current needs.
  - Cognitive: Depends on human perception and behavior.
  - Dynamic: Changes over time.

Human Labeled Corpora (Gold Standard)

- Start with a corpus of documents.
- Collect a set of queries for this corpus.
- Have one or more human experts exhaustively label the relevant documents for each query.
- Typically assumes binary relevance judgments.
- Requires considerable human effort for large document/query corpora.

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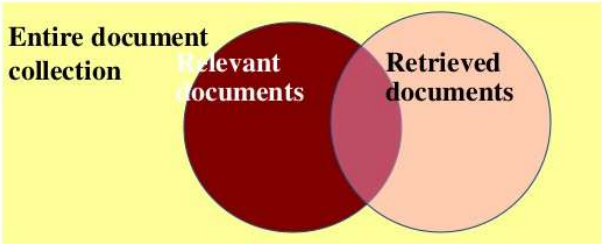
Precision and Recall

- Precision
  - The ability to retrieve top-ranked documents that are mostly relevant.
- Recall
  - The ability of the search to find all of the relevant items in the corpus.



	retrieved	not retrieved
irrelevant	retrieved & irrelevant	Not retrieved & irrelevant
relevant	retrieved & relevant	not retrieved but relevant

$$\text{precision} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of documents retrieved}}$$

$$\text{recall} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of relevant documents}}$$

Determining Recall is Difficult

- Total number of relevant items is sometimes not available:
  - Sample across the database and perform relevance judgment on these items.
  - Apply different retrieval algorithms to the same database for the same query. The aggregate of relevant items is taken as the total relevant set.

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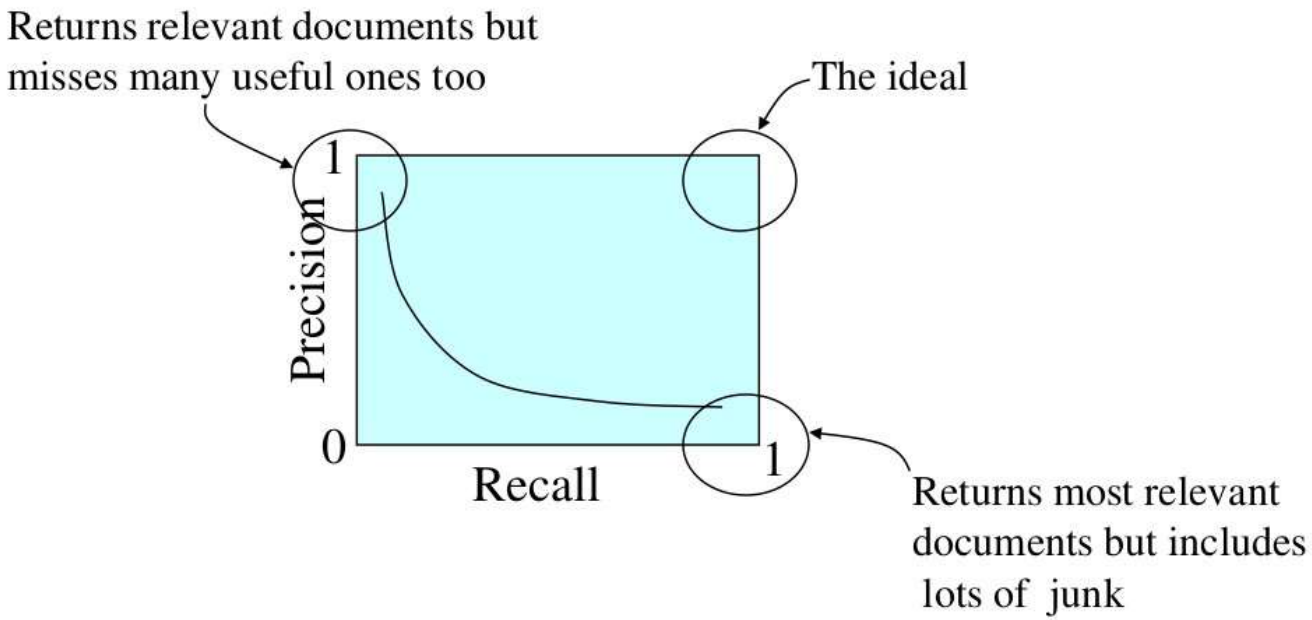
Trade-off between Recall and Precision



- The ideal
- Returns relevant documents but misses many useful ones too
- Returns most relevant documents but includes lots of junk

Computing Recall/Precision Points

- For a given query, produce the ranked list of retrievals.
- Adjusting a threshold on this ranked list produces different sets of retrieved documents, and therefore different recall/precision measures.
- Mark each document in the ranked list that is relevant according to the gold standard.
- Compute a recall/precision pair for each position in the ranked list that contains a relevant document.

Computing Recall/Precision Points: Example 1

n	doc #	relevant
1	588	x
2	589	x
3	576	
4	590	x
5	986	
6	592	x
7	984	
8	988	
9	578	
10	985	
11	103	
12	591	
13	772	x
14	990	

Let total # of relevant docs = 6
Check each new recall point:

R=1/6=0.167; P=1/1=1

R=2/6=0.333; P=2/2=1

R=3/6=0.5; P=3/4=0.75

R=4/6=0.667; P=4/6=0.667

R=5/6=0.833; p=5/13=0.38

Missing one relevant docu
Never reach 100% recall

Computing Recall/Precision Points: Example 2

n	doc #	relevant
1	588	x
2	576	
3	589	x
4	342	
5	590	x
6	717	
7	984	
8	772	x
9	321	x
10	498	
11	113	
12	628	
13	772	
14	592	x

Let total # of relevant docs = 6
Check each new recall point:

R=1/6=0.167; P=1/1=1

R=2/6=0.333; P=2/3=0.667

R=3/6=0.5; P=3/5=0.6

R=4/6=0.667; P=4/8=0.5

R=5/6=0.833; P=5/9=0.556

R=6/6=1.0; p=6/14=0.429

Average Recall/Precision Curve

- Typically average performance over a large set of queries.
- Compute average precision at each standard recall level across all queries.
- Plot average precision/recall curves to evaluate overall system performance on a document/query corpus.

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F1-Measure

- One measure of performance that takes into account both recall and precision.
- Harmonic mean of recall and precision:

$$F1 = \frac{2PR}{P+R} = \frac{2}{\frac{1}{P} + \frac{1}{R}}$$

- Compared to arithmetic mean, both need to be high for harmonic mean to be high.

F Measure

A variant of F measure that allows weighting emphasis on precision over recall:

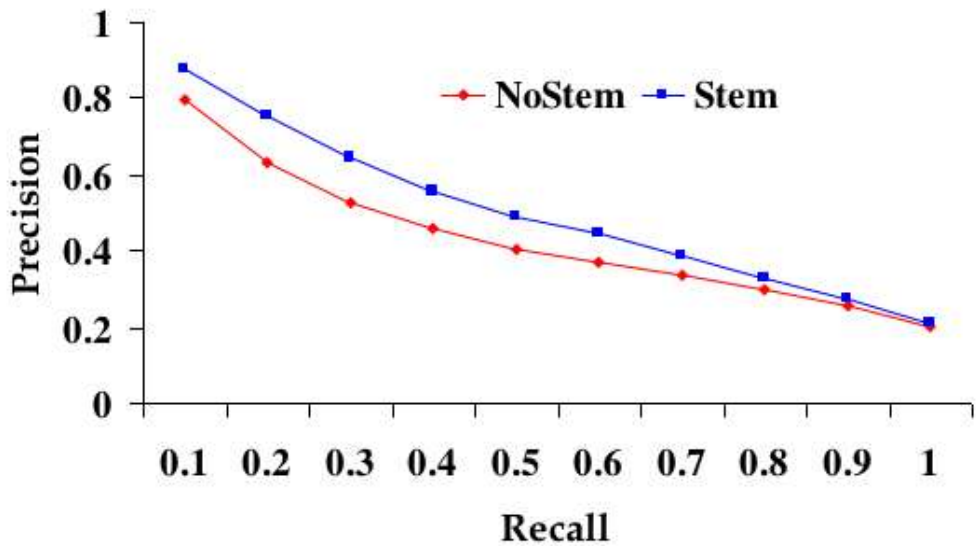
$$F = \frac{1}{a\frac{1}{P} + (1-a)\frac{1}{R}} = \frac{(\beta^2+1)PR}{\beta^2 P + R}$$

Value of β controls trade-off:

- $\beta = 1$: Equally weight precision and recall (E=F).
- $\beta > 1$: Weight recall more.
- $\beta < 1$: Weight precision more.

Precision-Recall Curve

- Compare Two or More Systems
- The curve closest to the upper right-hand corner of the graph indicates the best performance



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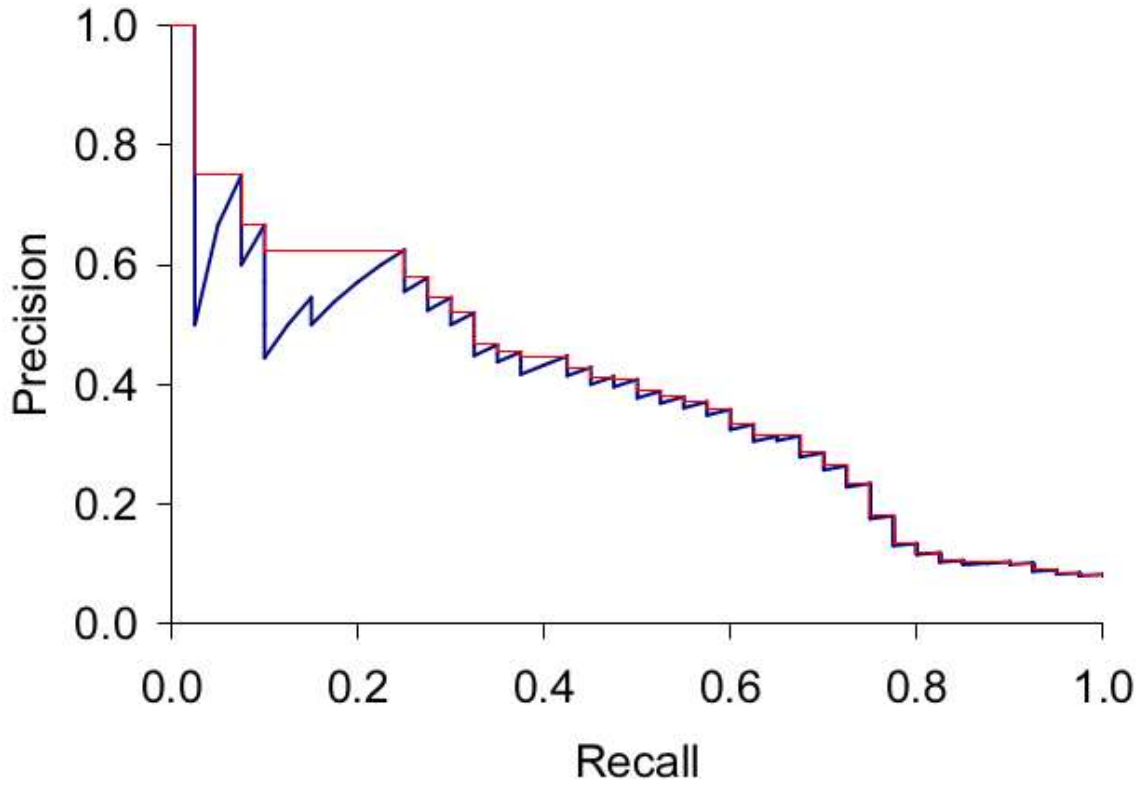
Rank-Based Measures

- Binary relevance
  - Precision@K (P@K)
  - Mean Average Precision (MAP)
- Multiple levels of relevance
  - Normalized Discounted Cumulative Gain (NDCG)

Precision-Recall Curve

- In a ranked retrieval context, appropriate sets of retrieved documents are given by the top k re-trieved documents.
- For each such set, precision and recall values can be plotted to give a precision-recall curve.
- Precision-recall curves have a distinctive saw-tooth shape.
  - If the (k + 1) th document retrieved is nonrelevant then recall is the same as for the top k documents, but precision has dropped.
  - If it is relevant, then both precision and recall increase, and the curve jags up and to the right.

Precision-Recall Curve



PRECISION@K

- Set a rank threshold K
- Compute % relevant in top K
- Ignores documents ranked lower than K
- Example: Green-relevant
  - Prec@3 of 2/3
  - Prec@4 of 2/4
  - Prec@5 of 3/5
- In similar fashion we have Recall@K

Average Precision

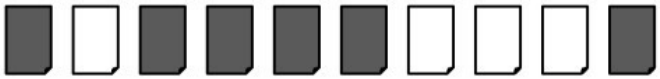
- Consider rank position of each relevant doc
- $K_1, K_2, \ldots K_R$
- Compute Precision@K for each $K_1, K_2, \ldots K_R$
- Average precision = average of P@K
- Example

$$\text{average precision} = \frac{1}{3}\ \frac{1}{1} + \frac{2}{3} + \frac{3}{5} = 0.76$$

Average Precision

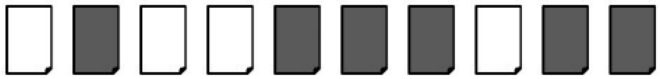
= the relevant documents

Ranking #1



Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6

Ranking #2



Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

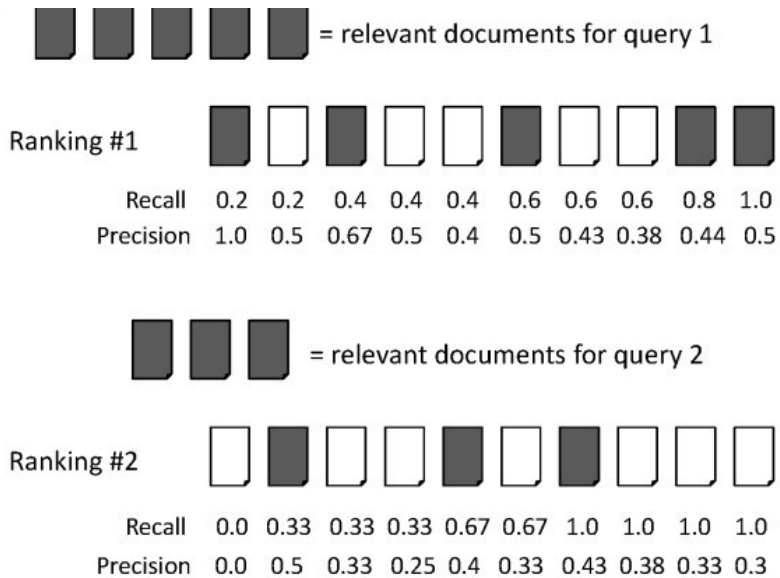
$$\text{Ranking \#1: } (1.0 + 0.67 + 0.75 + 0.8 + 0.83 + 0.6)/6 = 0.78$$

$$\text{Ranking \#2: } (0.5 + 0.4 + 0.5 + 0.57 + 0.56 + 0.6)/6 = 0.52$$

Mean Average Precision

- If a relevant document never gets retrieved, we assume the precision corresponding to that relevant doc to be zero
- MAP is macro-averaging: each query counts equally
- Now perhaps most commonly used measure in research papers
- MAP assumes user is interested in finding many relevant documents for each query
- MAP requires many relevance judgments in text collection

Mean Average Precision (MAP)



$$average\ precision\ query\ 1 = (1.0 + 0.67 + 0.5 + 0.44 + 0.5)/5 = 0.62$$
$$average\ precision\ query\ 2 = (0.5 + 0.4 + 0.43)/3 = 0.44$$

$$mean\ average\ precision = (0.62 + 0.44)/2 = 0.53$$

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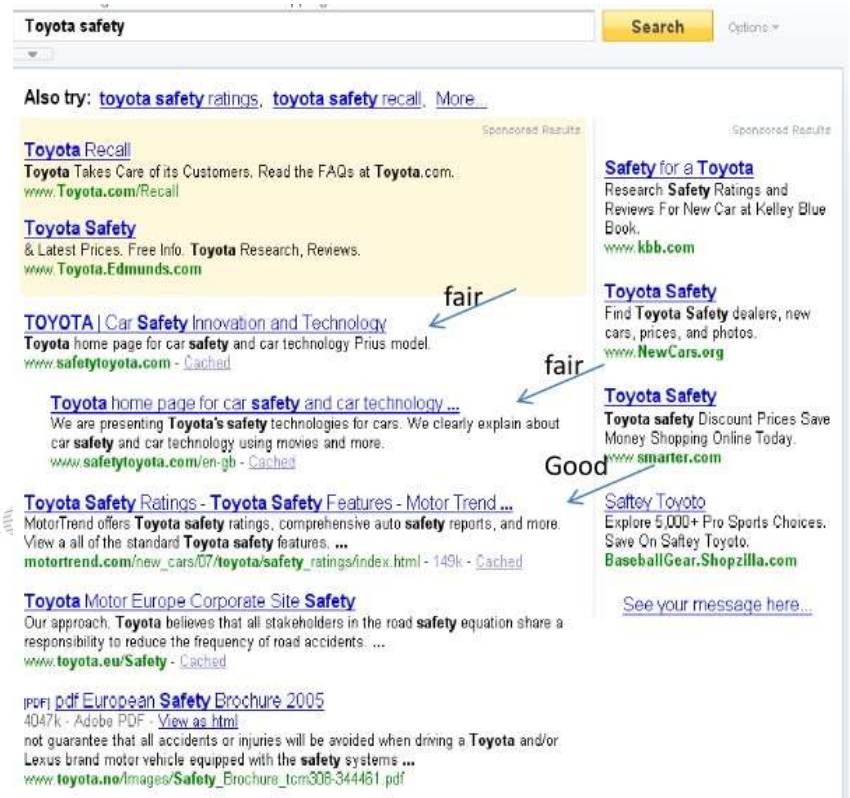
Non-Binary Relevance



Non-Binary Relevance

- Documents are rarely entirely relevant or non-relevant to a query
- Many sources of graded relevance judgments
  - Relevance judgments on a 5-point scale
  - Multiple judges
  - Click distribution and deviation from expected levels (but click-through != relevance judgments)

Discounted Cumulative Gain

- Popular measure for evaluating web search and related tasks
- Two assumptions:
  - Highly relevant documents are more useful than marginally relevant documents
  - The lower the ranked position of a relevant document, the less useful it is for the user, since it is less likely to be examined

Cumulative Gain

- With graded relevance judgments, we can compute the gain at each rank.
- Cumulative Gain at rank *n*:

$$CG_n = \sum_{i=1}^{n} rel_i$$

(Where rel_i is the graded relevance of the document at position i)

n	doc #	relevance (gain)	CG_n
1	588	1.0	1.0
2	589	0.6	1.6
3	576	0.0	1.6
4	590	0.8	2.4
5	986	0.0	2.4
6	592	1.0	3.4
7	984	0.0	3.4
8	988	0.0	3.4
9	578	0.0	3.4
10	985	0.0	3.4
11	103	0.0	3.4
12	591	0.0	3.4
13	772	0.2	3.6
14	990	0.0	3.6

Discounted Cumulative Gain (DCG)

- What if relevance judgments are in a scale of $[0, r]$? $r > 2$
- Cumulative Gain (CG) at rank n
  - Let the ratings of the n documents be $r_1, r_2, \ldots r_n$ (in ranked order)
  - $CG = r_1 + r_2 + \ldots r_n$
- Discounted Cumulative Gain (DCG) at rank n
  - $DCG = r_1 + r_2 / \log_2 2 + r_3 / \log_2 3 + \ldots r_n / \log_2 n$

$$DCG = rel_1 + \sum_{i=2}^{n} \frac{rel_i}{\log_2 i}$$

  - We may use any base for the logarithm
  - Summarize ranking

Discounting Based on Position

- Users care more about high-ranked documents, so we discount results by $1/\log_2(\text{rank})$
- Discounted Cumulative Gain:

$$DCG_n = rel_1 + \sum_{i=2}^{n} \frac{rel_i}{\log_2 i}$$

n	doc #	rel (gain)	CG_n	$\log_n$	DCG_n
1	588	1.0	1.0	-	1.00
2	589	0.6	1.6	1.00	1.60
3	576	0.0	1.6	1.58	1.60
4	590	0.8	2.4	2.00	2.00
5	986	0.0	2.4	2.32	2.00
6	592	1.0	3.4	2.58	2.39
7	984	0.0	3.4	2.81	2.39
8	988	0.0	3.4	3.00	2.39
9	578	0.0	3.4	3.17	2.39
10	985	0.0	3.4	3.32	2.39
11	103	0.0	3.4	3.46	2.39
12	591	0.0	3.4	3.58	2.39
13	772	0.2	3.6	3.70	2.44
14	990	0.0	3.6	3.81	2.44

Normalized Discounted Cumulative Gain

- To compare DCGs, normalize values so that a ideal ranking would have a Normalized DCG of 1.0

n	doc #	rel (gain)	CG_n	log_n	DCG_n
1	588	1.0	1.0	0.00	1.00
2	589	0.6	1.6	1.00	1.60
3	576	0.0	1.6	1.58	1.60
4	590	0.8	2.4	2.00	2.00
5	986	0.0	2.4	2.32	2.00
6	592	1.0	3.4	2.58	2.39
7	984	0.0	3.4	2.81	2.39
8	988	0.0	3.4	3.00	2.39
9	578	0.0	3.4	3.17	2.39
10	985	0.0	3.4	3.32	2.39
11	103	0.0	3.4	3.46	2.39
12	591	0.0	3.4	3.58	2.39
13	772	0.2	3.6	3.70	2.44
14	990	0.0	3.6	3.81	2.44

n	doc #	rel (gain)	CG_n	log_n	$IDCG_n$
1	588	1.0	1.0	0.00	1.00
2	592	1.0	2.0	1.00	2.00
3	590	0.8	2.8	1.58	2.50
4	589	0.6	3.4	2.00	2.80
5	772	0.2	3.6	2.32	2.89
6	576	0.0	3.6	2.58	2.89
7	986	0.0	3.6	2.81	2.89
8	984	0.0	3.6	3.00	2.89
9	988	0.0	3.6	3.17	2.89
10	578	0.0	3.6	3.32	2.89
11	985	0.0	3.6	3.46	2.89
12	103	0.0	3.6	3.58	2.89
13	591	0.0	3.6	3.70	2.89
14	990	0.0	3.6	3.81	2.89

NDCG for Summarizing Rankings

- Normalized Discounted Cumulative Gain (NDCG) at rank *n*
  - Normalize DCG at rank *n* by the DCG value at rank *n* of the ideal ranking
  - The ideal ranking would first return the documents with the highest relevance level, then the next highest relevance level, etc
- Normalization useful for contrasting queries with varying numbers of relevant results
- NDCG is now quite popular in evaluating Web search

Normalized Discounted Cumulative Gain (NDCG)

- Normalize by DCG of the ideal ranking:

$$NDCG = \frac{DCG_n}{IDCG_n}$$

- NDCG ≤ 1 at all ranks
- NDCG is comparable across different queries

n	doc #	rel (gain)	DCG_n	$IDCG_n$	$NDCG_n$
1	588	1.0	1.00	1.00	**1.00**
2	589	0.6	1.60	2.00	**0.80**
3	576	0.0	1.60	2.50	**0.64**
4	590	0.8	2.00	2.80	**0.71**
5	986	0.0	2.00	2.89	**0.69**
6	592	1.0	2.39	2.89	**0.83**
7	984	0.0	2.39	2.89	**0.83**
8	988	0.0	2.39	2.89	**0.83**
9	578	0.0	2.39	2.89	**0.83**
10	985	0.0	2.39	2.89	**0.83**
11	103	0.0	2.39	2.89	**0.83**
12	591	0.0	2.39	2.89	**0.83**
13	772	0.2	2.44	2.89	**0.84**
14	990	0.0	2.44	2.89	**0.84**

NDCG Summary

- Discounted Cumulative Gain (DCG, nDCG)
  - Sometimes relevance comes in more than one shade, e.g.

    0 = not relevant, 1 = somewhat rel, 2 = very relevant
  - There should be a "discount" in the score if very relevant documents come after only somewhat relevant documents
  - **Cumulative gain:** $CG@k = \Sigma_{i=1..k}\ rel_i$
  - **Discounted CG:** $DCG@k = rel_1 + \Sigma_{i=2..k}\ rel_i / \log_2 i$
  - Problem: CG and DCG are larger for larger result lists
  - Solution: normalize by maximally achievable value
  - **Ideal DCG:** iDCG@k = DCG@k of ideal ranking

NDCG Example

- Discounted Cumulative Gain (DCG, nDCG), example
  - Consider the following result list and relevances, assuming only 3 relevant documents overall (for this query)

Hit #1:	very relevant	2
Hit #2:	relevant	1
Hit #3:	not relevant	0
Hit #4:	very relevant	2
Hit #5:	not relevant	0

  - Then **DCG@5** = $2 + \frac{1}{\log_2 2} + \frac{2}{\log_2 4} = 2 + 1 + 1 =$

the „discount" for position 4

  - And **iDCG@5** = $2 + \frac{2}{\log_2 2} + \frac{1}{\log_2 3} = 2 + 2 + 0.63$
  - Hence **nDGC@5** = $4 / 4.63 = 0.86 = 86\%$

NDCG – Example 2

i	Ground Truth Document Order	r_i	Ranking Function 1 Document Order	r_i	Ranking Function 2 Document Order	r_i
1	d4	2	d3	2	d3	2
2	d3	2	d4	2	d2	1
3	d2	1	d2	1	d4	2
4	d1	0	d1	0	d1	0

$$DCG_{GT} = 2 + \frac{2}{\log_2 2} + \frac{1}{\log_2 3} + \frac{0}{\log_2 4} = 4.6309$$

$$DCG_{RF1} = 2 + \frac{2}{\log_2 2} + \frac{1}{\log_2 3} + \frac{0}{\log_2 4} = 4.6309$$

$$DCG_{RF2} = 2 + \frac{1}{\log_2 2} + \frac{2}{\log_2 3} + \frac{0}{\log_2 4} = 4.2619$$

$$MaxDCG = DCG_{GT} = 4.6309$$

SUMMARY

- Benchmarks consist of
  - Document collection
  - Query set
  - Assessment methodology
- Assessment methodology can use raters, user clicks, or a combination
- These get quantized into a goodness measure – Precision, Recall, P@K, MAP, NDCG.
- Different engines/algorithms compared on a benchmark together with a goodness measure

References

1. Chapter 8
   C. D. Manning, P. Raghavan and H. Schutze. Introduction to Information Retrieval. 2008. http://nlp.stanford.edu/IR-book/

Thank You!